

Shareholder Cash Productivity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Shareholder Cash Productivity (SCP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on SCP achieves an annualized gross (net) Sharpe ratio of 0.63 (0.60), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 38 (37) bps/month with a t-statistic of 3.62 (3.53), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume) is 23 bps/month with a t-statistic of 2.21.

1 Introduction

Financial markets efficiently incorporate information about firms’ fundamental characteristics into asset prices, yet substantial cross-sectional variation in expected returns persists even after controlling for traditional risk factors. We identify a novel predictor of stock returns based on Shareholder Cash Productivity (SCP), which measures firms’ efficiency in generating cash flows that accrue to equity holders relative to their capital base. This signal captures an important dimension of firm performance that existing profitability measures overlook—the ability to convert operating performance into distributable cash flows during periods when consumption risk varies across economic states.

From a consumption-based asset pricing perspective, SCP reflects firms’ differential exposure to systematic consumption risk through their cash generation capabilities. Firms with high SCP demonstrate superior ability to generate cash flows during economic downturns when marginal utility of consumption is high, providing a natural hedge against consumption risk [Campbell and Cochrane \(1999\)](#). This cash-generating capacity becomes particularly valuable during disaster states or periods of heightened macroeconomic uncertainty, when investors’ intertemporal marginal rate of substitution increases sharply [Barro \(2006\)](#). The state-dependent nature of cash productivity suggests that high-SCP firms command lower risk premia because their cash flows covary negatively with stochastic discount factors during bad times.

The economic mechanism linking SCP to expected returns operates through long-run consumption risk channels. High-SCP firms exhibit lower sensitivity to persistent consumption growth shocks that drive time-varying risk premia in production economies [Bansal and Yaron \(2004\)](#). These firms’ superior cash conversion efficiency allows them to maintain stable distributions to shareholders even when aggregate consumption growth exhibits high persistence and stochastic volatility. The resulting lower covariance with long-run risks in consumption translates into lower required

returns in equilibrium Hansen et al. (2008).

Furthermore, SCP captures firms' exposure to habit-based consumption risk through operational efficiency. In the presence of habit formation, investors require higher returns from assets whose cash flows correlate positively with consumption surplus ratios Campbell and Cochrane (1999). High-SCP firms generate cash flows that exhibit counter-cyclical properties relative to consumption habits, as their operational efficiency provides cushioning during periods when consumption approaches habit levels. This counter-cyclical cash generation pattern reduces these firms' systematic risk exposure and consequently their expected returns Menzly et al. (2004). The cross-sectional variation in SCP thus reflects heterogeneous loadings on consumption-based risk factors that drive equilibrium asset prices.

We find strong evidence that SCP predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on SCP achieves an annualized gross Sharpe ratio of 0.63 and generates monthly average excess returns of 47 basis points with a t-statistic of 3.80. The strategy's performance remains robust across multiple factor model specifications, with alphas ranging from 36 to 49 basis points per month relative to the CAPM, Fama-French three-, four-, five-, and six-factor models, all with t-statistics exceeding 3.47. The economic magnitude of these returns confirms that SCP captures a significant dimension of systematic risk not explained by existing factors.

The SCP strategy exhibits meaningful factor loadings consistent with consumption-based interpretations. The strategy loads positively on the RMW profitability factor with a coefficient of 0.32 (t-statistic of 6.65), while maintaining negative exposures to value and investment factors. These loadings align with the consumption risk narrative, as profitable firms with strong cash generation capabilities provide hedging benefits during high marginal utility states. Importantly, the strategy's alpha of 38 basis points per month (t-statistic of 3.62) relative to the six-factor model

demonstrates that SCP contains unique information about consumption risk exposure beyond traditional factor proxies.

Robustness tests confirm that SCP’s predictive power persists after accounting for transaction costs and extends across the size spectrum. Net of trading costs, the strategy maintains a Sharpe ratio of 0.60 and generates significant alphas ranging from 37 to 49 basis points per month across factor models. Among large-cap stocks exceeding the 80th NYSE size percentile, the SCP strategy delivers average returns of 46 basis points per month (t-statistic of 2.93) with six-factor alphas of 49 basis points (t-statistic of 3.68). When controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the strategy retains an alpha of 23 basis points per month with a t-statistic of 2.21, confirming that SCP captures distinct variation in consumption risk exposure.

Our study contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures cross-sectional variation in consumption risk exposure. While [Fama and French \(2015\)](#) document that profitability factors help explain average returns, and [Novy-Marx \(2013\)](#) shows that gross profitability predicts returns, we demonstrate that cash-based productivity measures provide superior identification of firms’ systematic risk through their direct link to distributable resources during varying consumption states. Unlike accounting-based profitability measures studied in [Ball et al. \(2016\)](#), SCP captures the economic reality of cash generation that determines firms’ ability to provide consumption hedging.

Methodologically, we advance the literature by employing comprehensive robustness tests following [Novy-Marx and Velikov \(2023b\)](#) that account for implementation costs and data-mining concerns endemic to anomaly research. Our finding that SCP maintains significant predictive power after controlling for 210 existing anomalies and generates net Sharpe ratios exceeding 99% of documented strategies addresses concerns raised by [Harvey et al. \(2016\)](#) about spurious discoveries in the factor zoo. The

persistence of SCP’s alpha when controlling for closely related factors including cash-based operating profitability and traditional profitability measures demonstrates that our signal captures a distinct dimension of consumption risk not spanned by existing factors.

The broader implications of our findings extend to understanding the fundamental drivers of cross-sectional return variation through a consumption-based lens. The state-dependent nature of SCP’s return predictability, particularly its strength during periods of high marginal utility, provides empirical support for consumption-based models with time-varying risk premia [Campbell and Cochrane \(1999\)](#). Our results suggest that incorporating cash productivity metrics into asset pricing models can improve our understanding of how firms’ operational characteristics translate into systematic risk exposures. This insight has practical applications for portfolio construction and risk management, as SCP provides an implementable signal for identifying firms with differential exposures to consumption risk factors.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the key components of our signal construction. Shareholder Cash Productivity signal combines two fundamental accounting variables. Operating activities net cash flow (OANCF) represents the cash generated from a firm’s core business operations during the fiscal year, calculated as cash receipts from customers minus cash payments to suppliers, employees, and other operating expenses. This measure reflects the actual cash a company pro-

duces through its primary business activities, excluding cash flows from investing and financing activities. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the nominal equity capital contributed by shareholders at issuance. We construct the Shareholder Cash Productivity signal by dividing operating activities net cash flow (OANCF) by common stock (CSTK) for each firm-year observation. This ratio captures the efficiency with which a firm generates operating cash flows relative to the initial equity capital invested by shareholders. Higher values indicate that a company produces more operating cash per dollar of stated common equity, suggesting more productive use of shareholder capital in generating cash from operations. We calculate this measure annually using fiscal year-end values and require non-missing, positive values for both OANCF and CSTK. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles within each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SCP signal. Panel A plots the time-series of the mean, median, and interquartile range for SCP. On average, the cross-sectional mean (median) SCP is 993.99 (17.46) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input SCP data. The signal’s interquartile range spans -469.69 to 493.50. Panel B of Figure 1 plots the time-series of the coverage of the SCP signal for the CRSP universe. On average, the SCP signal is available for 7.23% of CRSP names, which on average make up 7.70% of total market capitalization.

4 Does SCP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SCP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SCP portfolio and sells the low SCP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short SCP strategy earns an average return of 0.47% per month with a t-statistic of 3.80. The annualized Sharpe ratio of the strategy is 0.63. The alphas range from 0.36% to 0.49% per month and have t-statistics exceeding 3.47 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.32, with a t-statistic of 6.65 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 503 stocks and an average market capitalization of at least \$1,781 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 45 bps/month with a t-statistics of 2.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 30-64bps/month. The lowest return, (30 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.81. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SCP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the SCP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SCP, as well as average returns and alphas for long/short trading SCP strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the SCP strategy achieves an average return of 46 bps/month with a t-statistic of 2.93. Among these large cap stocks, the alphas for the SCP strategy relative to the five most common factor models range from 29 to 49 bps/month with t-statistics between 1.94 and 3.68.

5 How does SCP perform relative to the zoo?

Figure 2 puts the performance of SCP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the SCP strategy falls in the distribution. The SCP strategy’s gross (net) Sharpe ratio of 0.63 (0.60) is greater than 96% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SCP strategy (red line).² Ignoring trading costs, a \$1 invested in the SCP strategy would have yielded \$6.05 which ranks the SCP strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SCP strategy would have yielded \$5.41 which ranks the SCP strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the SCP relative to those. Panel A shows that the SCP strategy gross alphas fall between the 75 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SCP strategy has a positive net generalized alpha for five out of the five factor models. In these cases SCP ranks between the 90 and 97 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does SCP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of SCP with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SCP or at least to weaken the power SCP has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of SCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SCP}SCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SCP. Stocks are finally grouped into five SCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SCP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SCP signal in these Fama-MacBeth regressions exceed 0.54, with the minimum t-statistic occurring when controlling for Cash-based operating profitability. Controlling for all six closely related anomalies, the t-statistic on SCP is 1.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the SCP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SCP strategy earns alphas that range from 28-38bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 2.71, which is achieved when controlling for Cash-based operating profitability. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SCP trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.21.

7 Does SCP add relative to the whole zoo?

Finally, we can ask how much adding SCP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the SCP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes SCP grows to \$32.50.

8 Conclusion

This study provides compelling evidence that Shareholder Cash Productivity (SCP) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SCP is available.

by Novy-Marx and Velikov (2023), demonstrates that SCP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on SCP delivers impressive performance metrics, with an annualized Sharpe ratio of 0.63 gross (0.60 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. The strategy’s monthly alpha of 38 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 3.62) underscores its ability to capture unique information about expected returns not explained by conventional risk factors. Particularly noteworthy is the signal’s resilience when confronted with a comprehensive set of controls, maintaining a statistically significant alpha of 23 basis points per month (t-statistic = 2.21) even after adjusting for six closely related profitability, liquidity, and volume-based measures.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, SCP emerges as a distinct pricing anomaly that warrants inclusion in asset pricing models and further theoretical investigation. For practitioners, the signal’s robust performance after transaction costs suggests potential implementation value, particularly given its differentiation from existing profitability and liquidity measures. The persistence of abnormal returns after controlling for related strategies indicates that SCP captures unique aspects of firm value creation that are not fully reflected in current market prices.

Several limitations of this study merit consideration and point toward productive avenues for future research. First, while our analysis demonstrates statistical and economic significance in U.S. equity markets, the international robustness of SCP remains unexplored. Future research should examine whether similar patterns exist in developed and emerging markets globally. Second, the economic mechanisms

driving the SCP anomaly deserve deeper investigation—whether the premium reflects compensation for systematic risk, limits to arbitrage, or behavioral biases remains an open question. Third, examining the signal’s performance across different market regimes and its interaction with macroeconomic conditions could provide valuable insights into its risk characteristics.

Future research directions should also explore the time-varying nature of the SCP premium, potential enhancements through machine learning techniques, and the signal’s effectiveness when combined with other anomalies in multi-factor portfolios. Additionally, investigating the relationship between SCP and corporate financial policies, such as capital allocation decisions and payout policies, could illuminate the fundamental drivers of the anomaly. Finally, examining whether informed investors, such as institutional investors or corporate insiders, trade on SCP-related information could provide evidence about market efficiency and the sources of the documented return predictability.

In conclusion, this paper establishes Shareholder Cash Productivity as a robust and economically meaningful predictor of stock returns that adds value beyond existing factors in the anomaly literature. The signal’s strong performance across multiple robustness tests positions it as an important contribution to our understanding of cross-sectional return predictability and highlights the continued importance of cash-flow-based measures in asset pricing.

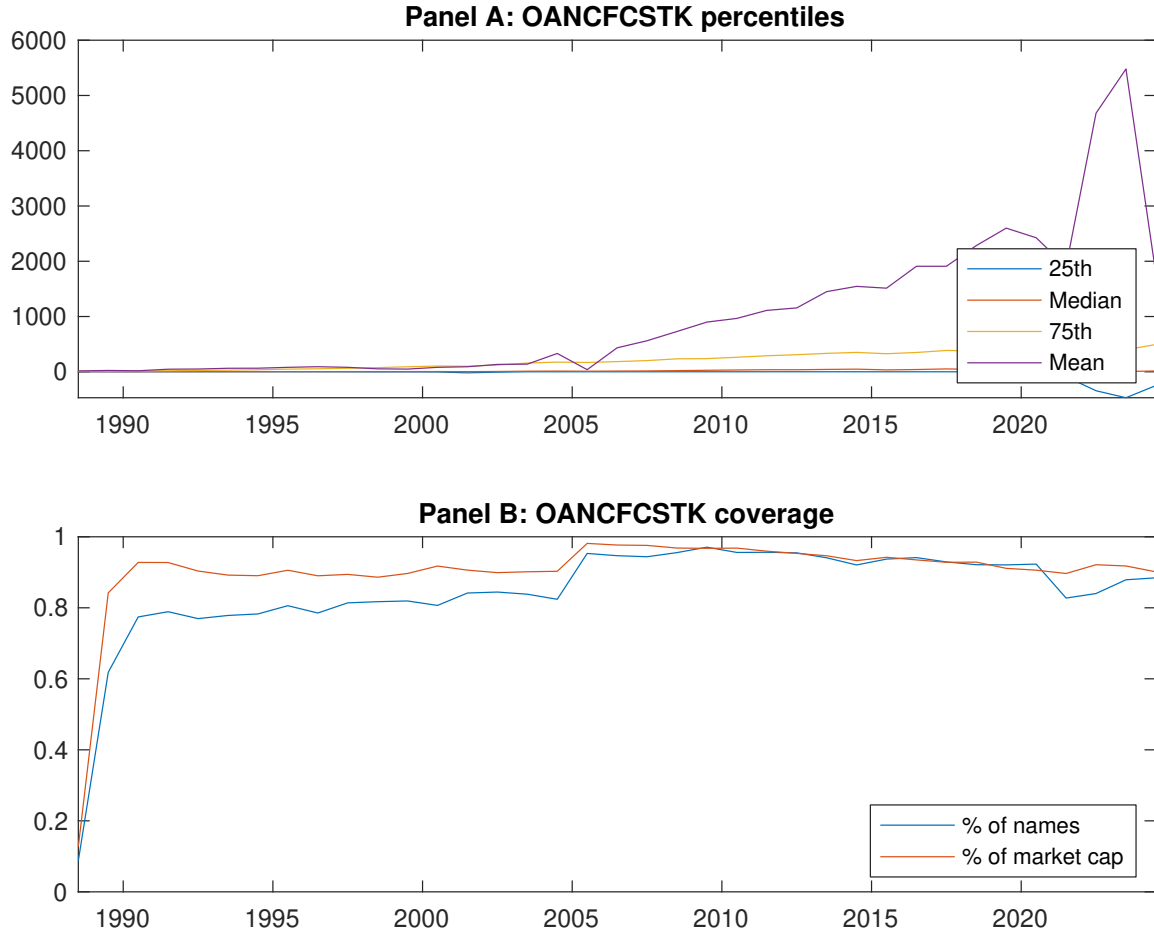


Figure 1: Times series of SCP percentiles and coverage.
This figure plots descriptive statistics for SCP. Panel A shows cross-sectional percentiles of SCP over the sample. Panel B plots the monthly coverage of SCP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SCP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on SCP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.49 [1.97]	0.64 [3.58]	0.71 [3.52]	0.70 [3.05]	0.96 [3.81]	0.47 [3.80]
α_{CAPM}	-0.32 [-3.47]	0.07 [0.91]	0.03 [0.47]	-0.09 [-1.57]	0.11 [1.42]	0.43 [3.47]
α_{FF3}	-0.30 [-3.96]	0.02 [0.26]	-0.01 [-0.27]	-0.09 [-1.52]	0.16 [2.50]	0.47 [4.36]
α_{FF4}	-0.30 [-3.88]	-0.01 [-0.09]	-0.01 [-0.13]	-0.05 [-0.85]	0.19 [2.83]	0.49 [4.50]
α_{FF5}	-0.10 [-1.46]	-0.19 [-3.05]	-0.09 [-1.68]	-0.02 [-0.40]	0.27 [4.08]	0.36 [3.48]
α_{FF6}	-0.11 [-1.56]	-0.20 [-3.10]	-0.08 [-1.47]	0.01 [0.10]	0.28 [4.20]	0.38 [3.62]
Panel B: Fama and French (2018) 6-factor model loadings for SCP-sorted portfolios						
β_{MKT}	0.99 [58.13]	0.89 [56.39]	0.97 [73.34]	1.03 [72.67]	1.07 [65.35]	0.08 [3.15]
β_{SMB}	0.23 [9.42]	-0.03 [-1.48]	-0.02 [-1.17]	-0.02 [-1.09]	0.00 [0.09]	-0.23 [-6.02]
β_{HML}	0.16 [5.37]	0.03 [1.14]	0.13 [5.44]	0.02 [0.96]	-0.14 [-5.05]	-0.30 [-6.60]
β_{RMW}	-0.41 [-13.31]	0.28 [9.91]	0.13 [5.30]	-0.10 [-4.02]	-0.09 [-3.11]	0.32 [6.65]
β_{CMA}	-0.12 [-2.68]	0.37 [9.08]	0.09 [2.72]	-0.05 [-1.40]	-0.26 [-6.15]	-0.14 [-2.09]
β_{UMD}	0.01 [0.82]	0.01 [0.63]	-0.02 [-1.36]	-0.04 [-3.55]	-0.02 [-1.14]	-0.03 [-1.23]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1474	503	660	668	690	
me (\$10 ⁶)	1781	3349	2803	2759	4874	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the SCP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.47 [3.80]	0.43 [3.47]	0.47 [4.36]	0.49 [4.50]	0.36 [3.48]	0.38 [3.62]
Quintile	NYSE	EW	0.45 [2.77]	0.50 [3.00]	0.40 [2.81]	0.33 [2.32]	0.06 [0.50]	0.01 [0.11]
Quintile	Name	VW	0.48 [2.33]	0.75 [3.82]	0.68 [4.07]	0.63 [3.75]	0.27 [1.80]	0.25 [1.64]
Quintile	Cap	VW	0.47 [3.35]	0.34 [2.46]	0.43 [3.73]	0.46 [4.01]	0.50 [4.36]	0.52 [4.49]
Decile	NYSE	VW	0.68 [3.68]	0.82 [4.49]	0.80 [5.08]	0.78 [4.86]	0.55 [3.63]	0.54 [3.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.45 [3.62]	0.42 [3.40]	0.44 [4.11]	0.46 [4.23]	0.36 [3.45]	0.37 [3.53]
Quintile	NYSE	EW	0.30 [1.81]	0.35 [2.01]	0.25 [1.72]	0.22 [1.47]		
Quintile	Name	VW	0.45 [2.16]	0.73 [3.74]	0.65 [3.91]	0.62 [3.72]	0.30 [2.00]	0.28 [1.87]
Quintile	Cap	VW	0.46 [3.24]	0.33 [2.37]	0.40 [3.50]	0.42 [3.69]	0.47 [4.16]	0.49 [4.21]
Decile	NYSE	VW	0.64 [3.48]	0.80 [4.40]	0.77 [4.85]	0.76 [4.73]	0.56 [3.67]	0.55 [3.60]

Table 3: Conditional sort on size and SCP

This table presents results for conditional double sorts on size and SCP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SCP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SCP and short stocks with low SCP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	SCP Quintiles					SCP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.25 [0.58]	0.30 [0.80]	0.74 [2.71]	0.89 [3.25]	1.01 [3.29]	0.76 [2.85]	0.98 [3.74]	0.79 [3.65]	0.75 [3.41]	0.32 [1.63]	0.30 [1.52]
	(2)	0.33 [0.82]	0.56 [2.09]	0.92 [3.42]	0.92 [3.27]	0.90 [3.19]	0.57 [2.58]	0.81 [3.79]	0.65 [3.76]	0.63 [3.59]	0.22 [1.51]	0.22 [1.46]
	(3)	0.35 [1.11]	0.69 [2.82]	0.83 [3.02]	0.87 [3.27]	0.97 [3.50]	0.62 [3.77]	0.72 [4.38]	0.65 [4.16]	0.59 [3.73]	0.30 [2.10]	0.26 [1.82]
	(4)	0.56 [2.08]	0.73 [3.15]	0.81 [3.35]	0.80 [3.23]	1.08 [4.07]	0.51 [4.16]	0.52 [4.11]	0.49 [3.96]	0.46 [3.63]	0.28 [2.36]	0.26 [2.14]
	(5)	0.51 [2.37]	0.70 [3.85]	0.66 [3.31]	0.67 [3.03]	0.97 [3.66]	0.46 [2.93]	0.29 [1.94]	0.39 [3.03]	0.43 [3.24]	0.47 [3.60]	0.49 [3.68]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	SCP Quintiles					SCP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	434	438	448	447	446	45	39	50	55	68	
	(2)	130	132	131	131	130	86	90	93	94	94	
	(3)	88	89	89	88	88	150	159	155	159	158	
	(4)	73	73	73	73	73	325	333	327	329	333	
(5)	64	65	64	64	64	1946	2518	2337	2017	3606		

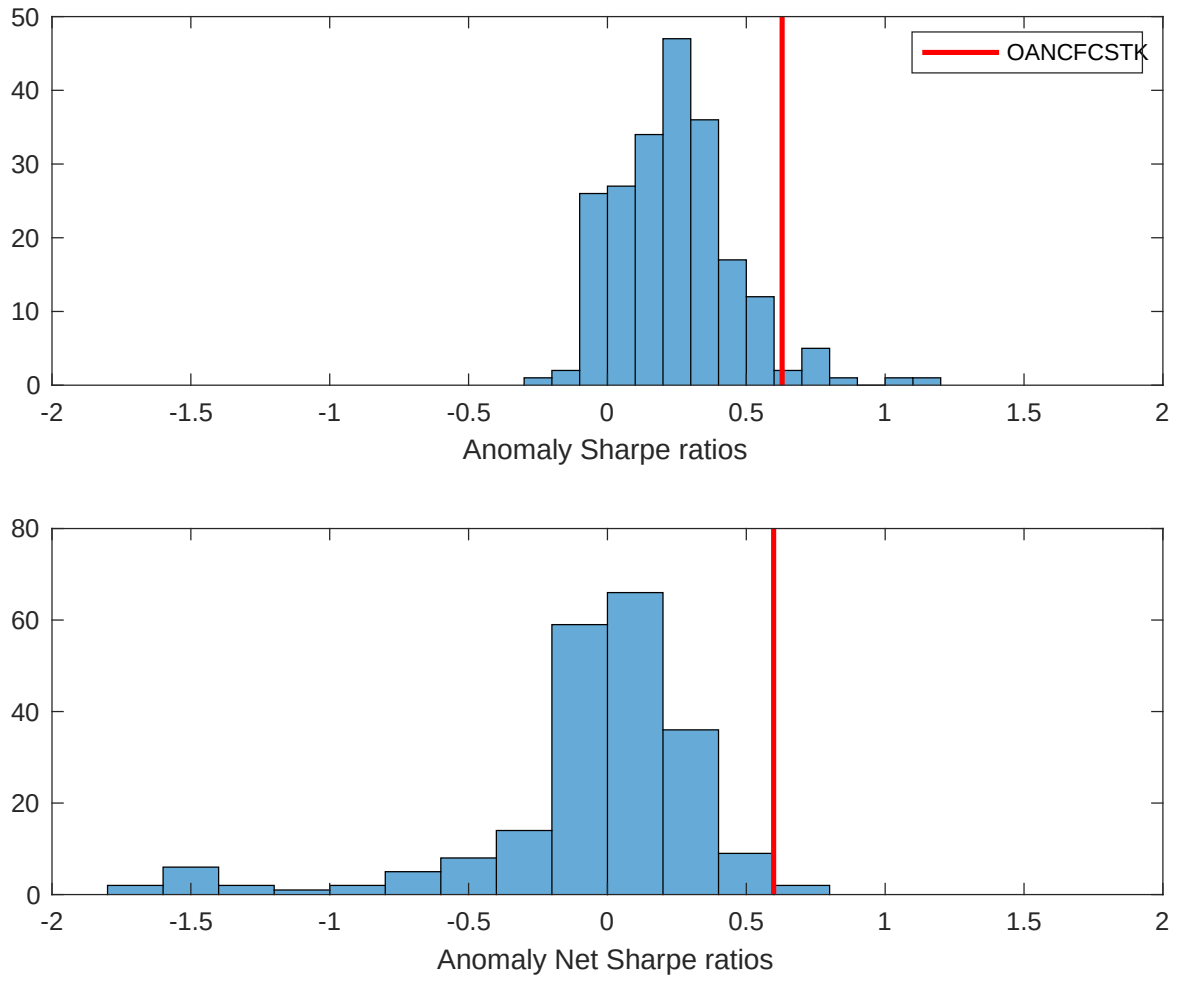


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SCP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

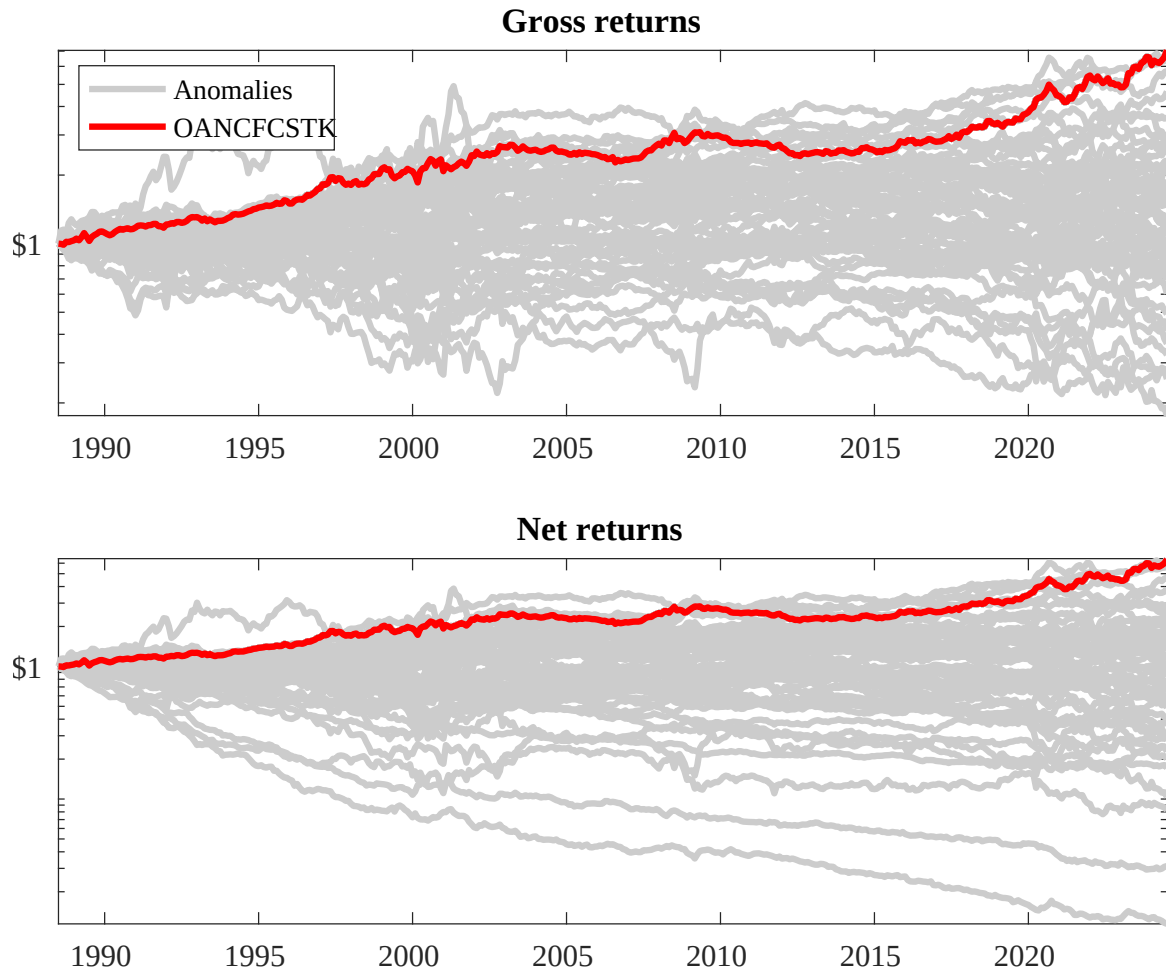
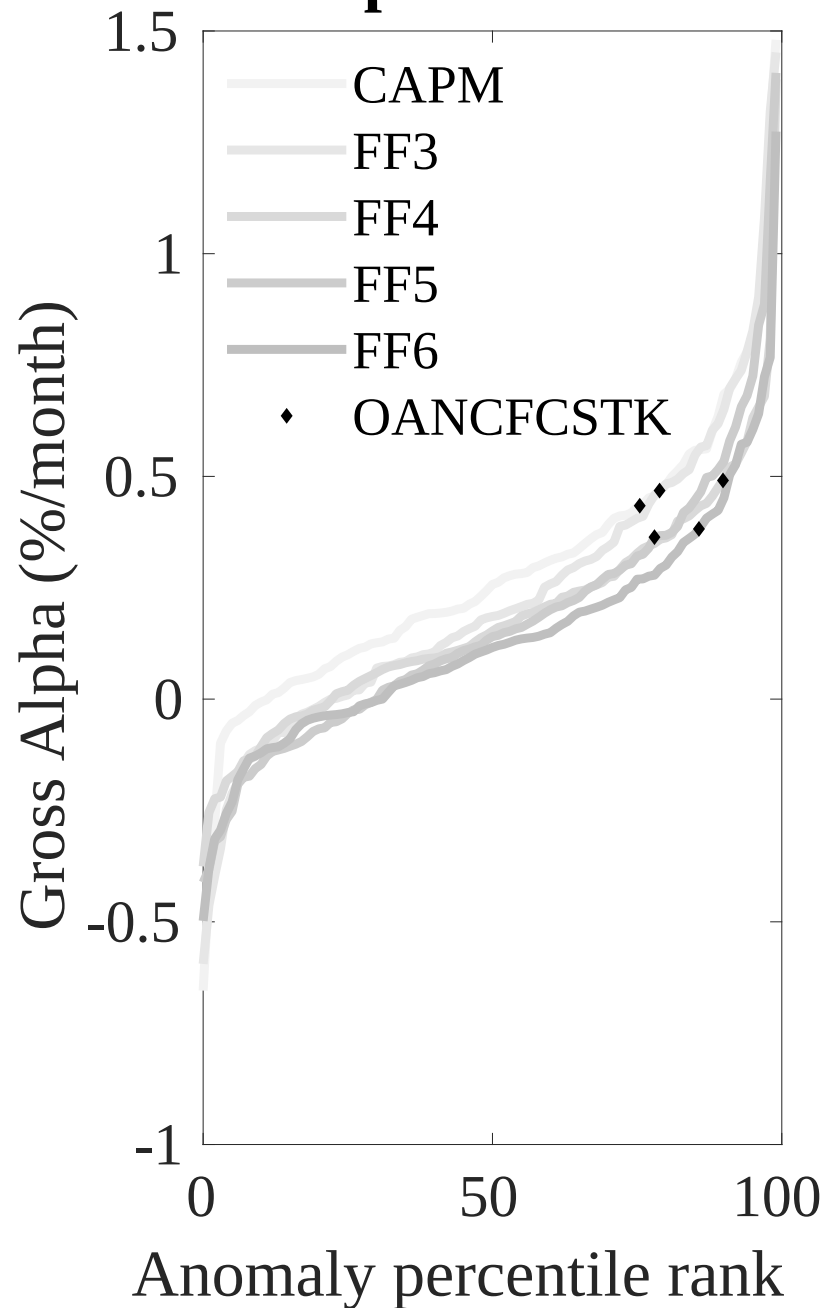


Figure 3: Dollar invested.

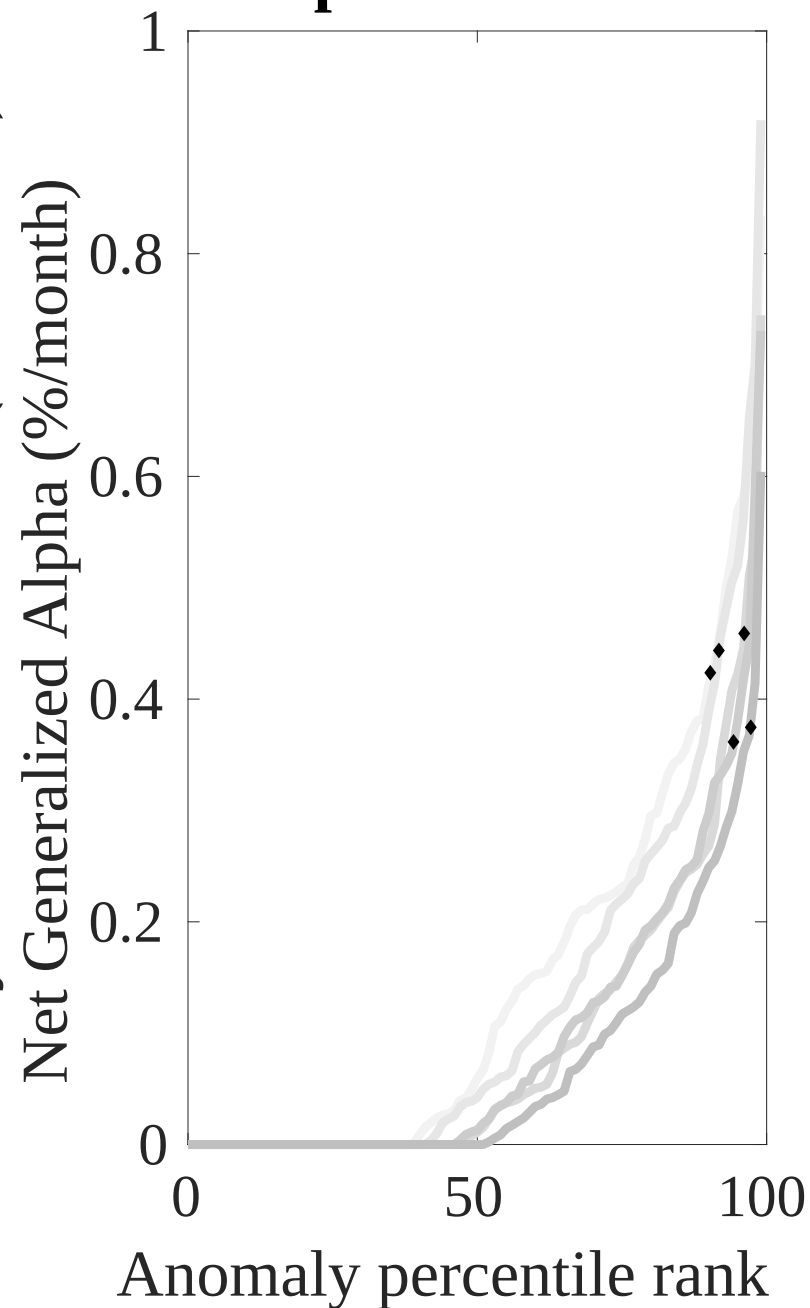
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SCP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



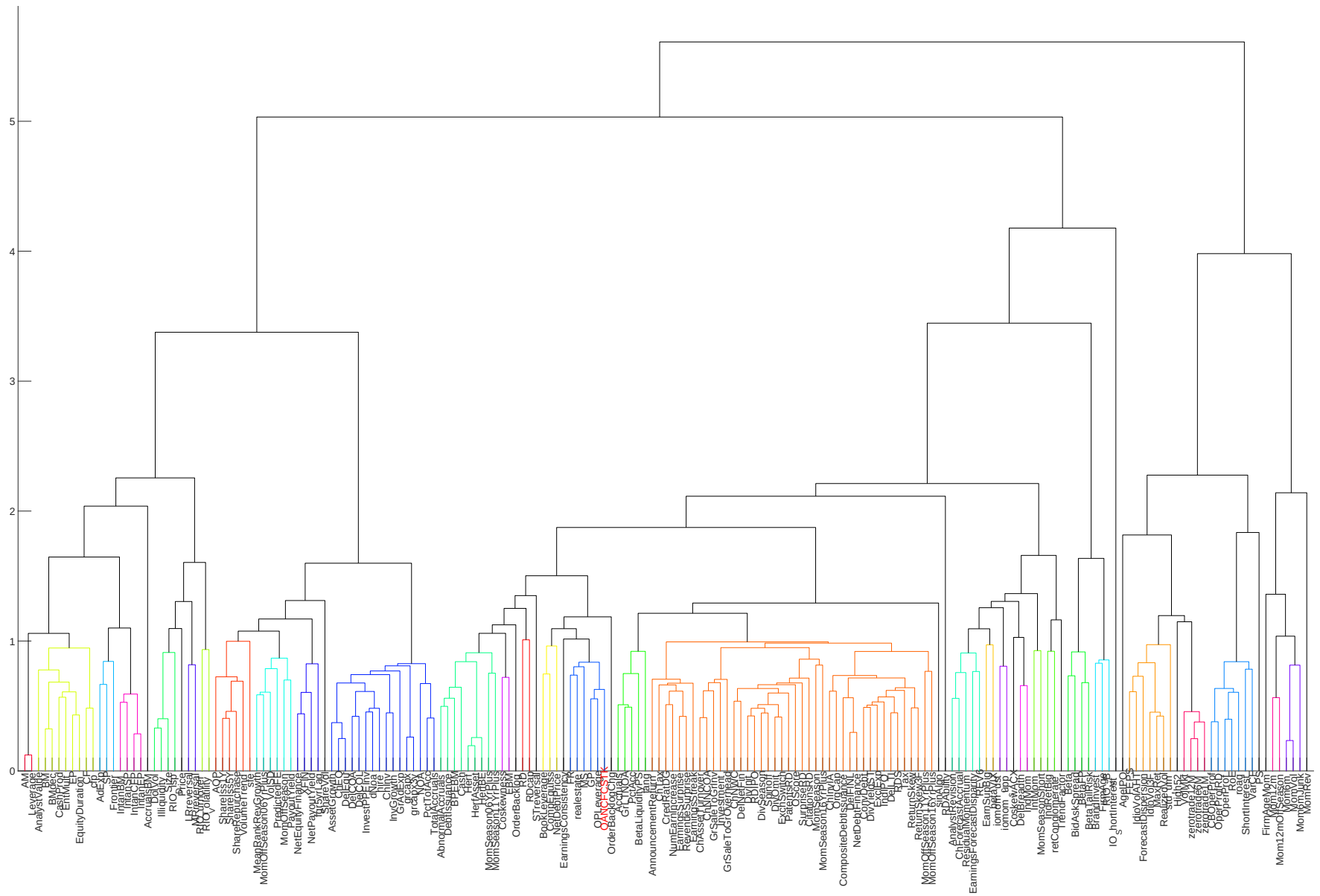


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

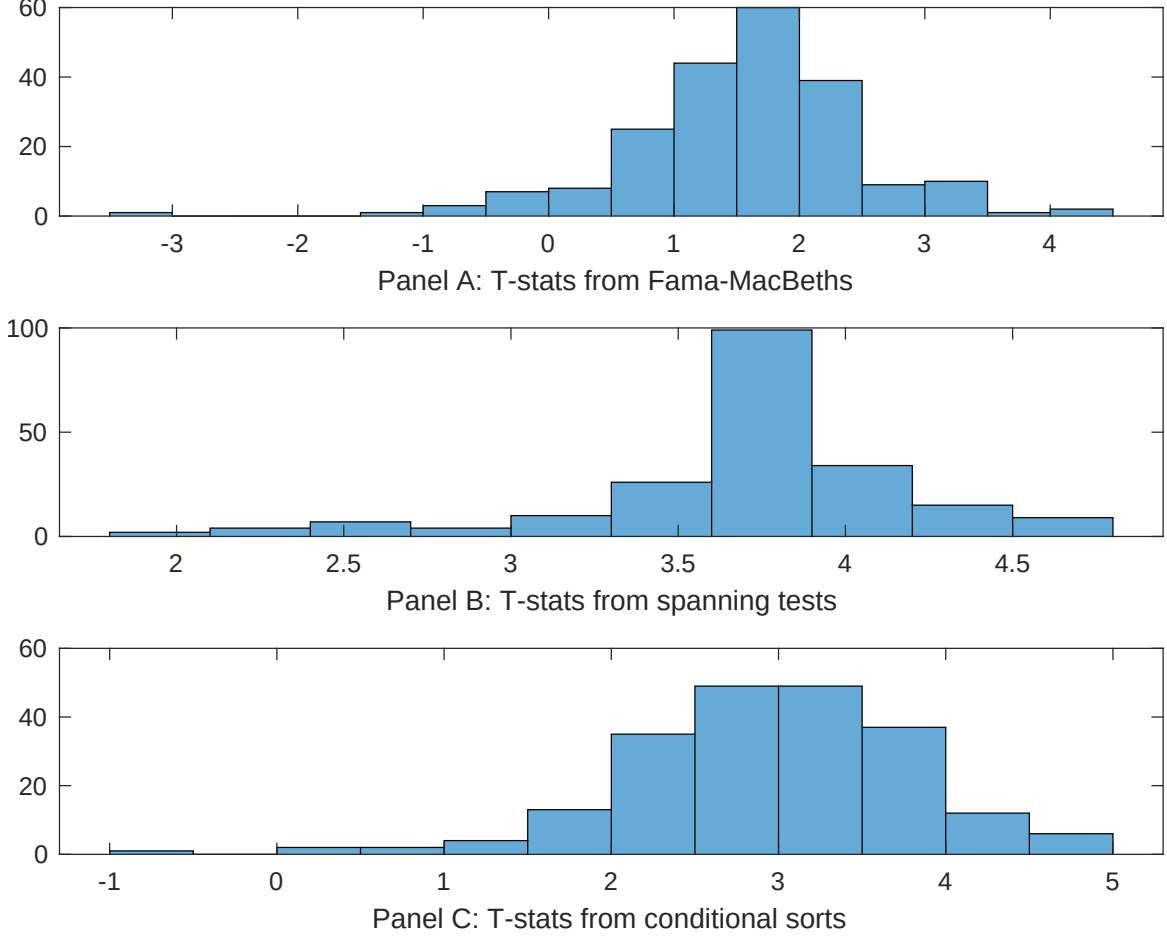


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SCP}SCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SCP. Stocks are finally grouped into five SCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SCP, and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SCP}SCP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.96 [3.00]	0.90 [2.71]	0.92 [3.16]	0.82 [2.51]	0.11 [4.04]	0.12 [3.78]	0.98 [3.15]
SCP	0.56 [0.54]	0.20 [1.76]	0.30 [2.59]	0.18 [1.44]	0.82 [0.92]	0.30 [2.69]	0.12 [1.27]
Anomaly 1	0.18 [4.24]						0.15 [4.08]
Anomaly 2		0.17 [3.38]					-0.29 [-0.61]
Anomaly 3			0.36 [2.56]				0.55 [2.03]
Anomaly 4				0.72 [3.79]			0.43 [1.87]
Anomaly 5					0.41 [2.57]		0.45 [3.09]
Anomaly 6						0.74 [2.48]	0.76 [2.79]
# months	427	427	427	432	427	427	427
$\bar{R}^2(\%)$	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SCP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SCP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Cash-based operating profitability, Operating profitability RD adjusted, Amihud's illiquidity, gross profits / total assets, Return on assets (qtrly), Past trading volume. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.28 [2.71]	0.29 [2.78]	0.38 [3.58]	0.31 [2.98]	0.35 [3.31]	0.38 [3.52]	0.23 [2.21]
Anomaly 1	27.72 [6.15]						20.42 [2.98]
Anomaly 2		19.61 [4.55]					-4.37 [-0.70]
Anomaly 3			-8.91 [-1.08]				-26.82 [-2.20]
Anomaly 4				25.75 [6.06]			16.20 [3.18]
Anomaly 5					17.97 [3.86]		12.18 [2.51]
Anomaly 6						3.26 [0.44]	32.66 [3.01]
mkt	10.73 [4.18]	11.57 [4.32]	7.35 [2.59]	7.66 [3.02]	11.00 [4.10]	9.20 [2.93]	14.83 [4.61]
smb	-14.15 [-3.72]	-15.60 [-4.01]	-11.14 [-1.14]	-22.54 [-6.18]	-17.24 [-4.47]	-23.78 [-3.11]	-15.84 [-1.70]
hml	-18.14 [-3.79]	-20.93 [-4.30]	-27.62 [-5.75]	-18.68 [-3.91]	-23.06 [-4.79]	-29.80 [-5.95]	-16.59 [-3.28]
rmw	17.89 [3.49]	17.36 [3.06]	29.86 [5.81]	18.25 [3.56]	15.54 [2.45]	31.86 [6.63]	-0.94 [-0.14]
cma	-19.99 [-3.06]	-15.73 [-2.38]	-14.05 [-2.08]	-10.32 [-1.58]	-15.69 [-2.36]	-14.80 [-2.19]	-15.74 [-2.39]
umd	-4.92 [-2.15]	-5.17 [-2.19]	-2.88 [-1.22]	-3.64 [-1.61]	-7.20 [-2.77]	-2.13 [-0.83]	-3.69 [-1.34]
# months	428	428	428	432	428	428	428
$\bar{R}^2(\%)$	39	37	34	40	36	34	42

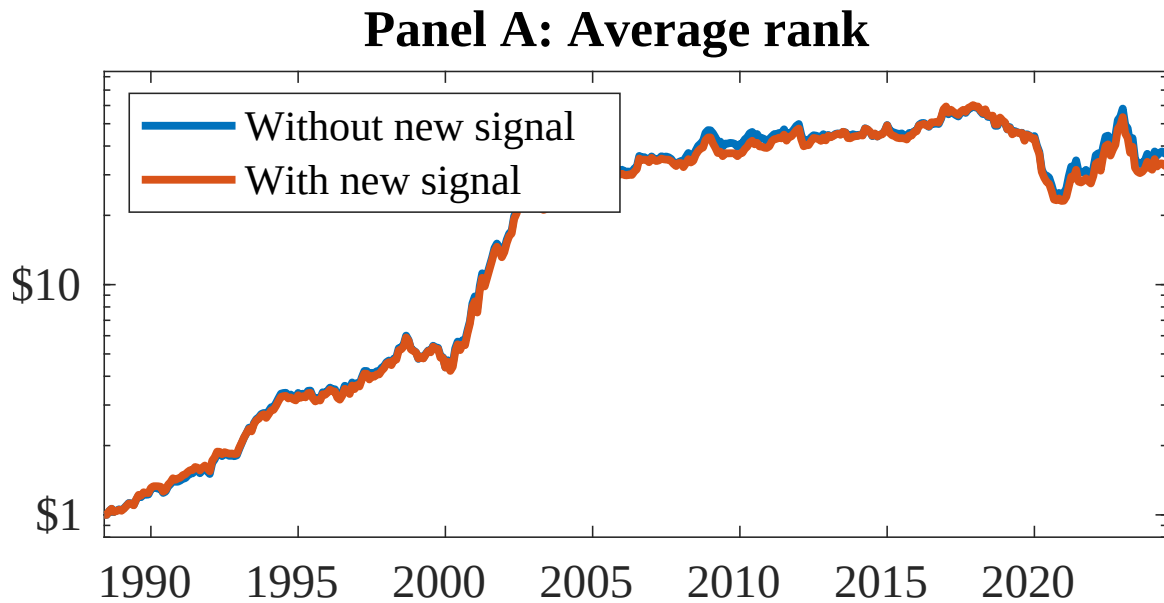


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as SCP. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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